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Answer (a) zero (b) zero (c) 0.6 x 10^{-23} N

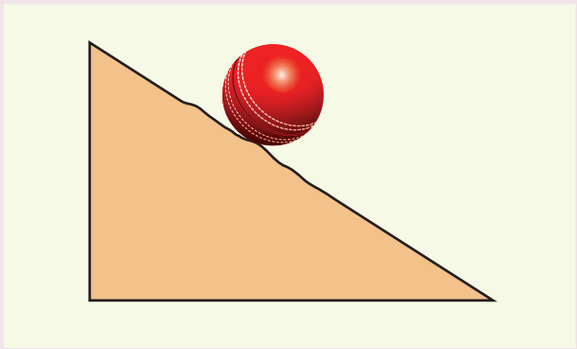
4. A bar magnet is placed in a uniform magnetic field whose strength is 0.8 T. Suppose the bar magnet orient at an angle 30° with the external field experiences a torque of 0.2 N m. Calculate:

(i) the magnetic moment of the magnet

(ii) the work done by an applied force in moving it from most stable configuration to the most unstable configuration and also compute the work done by the applied magnetic field in this case.

Answer (i) 0.5 A m^2 (ii) W = 0.8 J and W_{mag} = -0.8 J

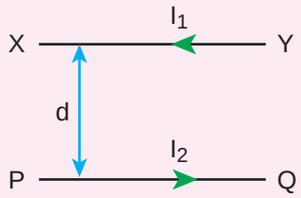
5. A non - conducting sphere has a mass of 100 g and radius 20 cm. A flat compact coil of wire with turns 5 is wrapped tightly around it with each turns concentric with the sphere. This sphere is placed on an inclined plane such that plane of coil is parallel to the inclined plane. A uniform magnetic field of 0.5 T exists in the region in vertically upward direction. Compute the current I required to rest the sphere in equilibrium. Answer $\frac{2}{\pi}A$



6. Calculate the magnetic field at the center of a square loop which carries a current of 1.5 A, length of each loop is 50 cm. Answer 3.4 x 10^{-6} T

7. Show that the magnetic field at any point on the axis of the solenoid having *n* turns per unit length is $B = \frac{1}{2}\mu_{\circ} nI(\cos\theta_1 - \cos\theta_2)$.

8. Let I_1 and I_2 be the steady currents passing through a long horizontal wire XY and PQ respectively. The wire PQ is fixed in horizontal plane and the wire XY be is allowed to move freely in a vertical plane. Let the wire XY is in equilibrium at a height *d* over the parallel wire PQ as shown in figure.



Show that if the wire XY is slightly displaced and released, it executes Simple Harmonic Motion (SHM). Also, compute the time period of oscillations.

Answer $a_y = -\omega^2 y$ (SHM) and time period

$$T = 2\pi\sqrt{\frac{d}{g}} \text{ in sec}$$

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